EXPERIMENT -4

AIM- Using the Employee schema, write queries with INNER JOIN, LEFT JOIN, self-join, 3-way join, correlated subqueries, EXISTS, and simulated INTERSECT and EXCEPT. Compare execution plans using EXPLAIN.

OBJECTIVES

1. To understand and implement different types of SQL joins.
2. To retrieve related data using INNER JOIN and LEFT JOIN.
3. To understand and implement a self-join on the Employee table.
4. To perform a 3-way join between Employee, Department, and Project tables.
5. To understand and implement correlated subqueries.
6. To use the EXISTS operator for checking the existence of related records.
7. To simulate the functionality of INTERSECT and EXCEPT using SQL queries.
8. To analyze and compare the execution plans of SQL queries using EXPLAIN.

THEORY

INNER JOIN

An INNER JOIN returns only those records that have matching values in both tables based on the specified join condition. It is used to combine related information from two or more tables.

For example, the Employee and Department tables can be joined using the common DeptID attribute to display employees along with their departments.

LEFT JOIN

A LEFT JOIN returns all records from the left table and the matching records from the right table. If no matching record exists in the right table, NULL values are returned for the columns of the right table.

It is useful when all records from the primary or left-side table must be retained, even if they do not have corresponding records in the other table.

SELF-JOIN

A self-join is a join in which a table is joined with itself. Different aliases are assigned to the same table so that its records can be compared with one another.

It can be used to find relationships between employees, such as employees working in the same department or employees having similar salaries.

3-WAY JOIN

A 3-way join combines three tables in a single query using related attributes.

In the Employee schema, the Employee table can be joined with both Department and Project tables. This allows information such as employee name, department name, and project name to be retrieved together.

CORRELATED SUBQUERY

A correlated subquery is a subquery that depends on a value from the outer query. Unlike an independent subquery, it is logically evaluated for each row considered by the outer query.

Correlated subqueries are useful when the result of the inner query needs to be compared with the current row of the outer query, such as finding employees whose salary is greater than the average salary of their own department.

EXISTS

The EXISTS operator checks whether a subquery returns at least one row.

It returns TRUE when the subquery produces one or more matching records and FALSE when no matching record exists. It is commonly used to determine whether a related record exists in another table.

SIMULATED INTERSECT-

INTERSECT returns records that are common to the results of two queries.

When INTERSECT is not supported by a particular SQL system, its functionality can be simulated using alternatives such as INNER JOIN or EXISTS. The columns and data types involved in the comparison should be compatible.

SIMULATED EXCEPT-

EXCEPT returns records that are present in the first query result but absent from the second query result.

When EXCEPT is not supported, it can be simulated using NOT EXISTS or an appropriate LEFT JOIN with a NULL check

EXPLAIN

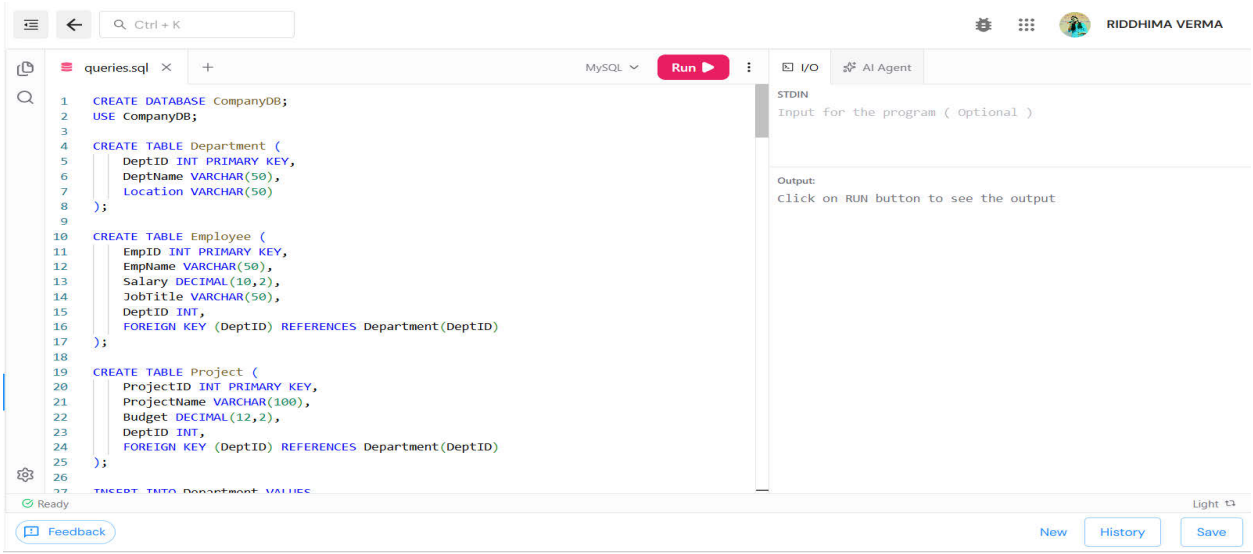
EXPLAIN is used to display the execution plan that the database optimizer intends to use for a SQL query.

It provides information about how tables will be accessed and joined, such as:

- Table scans
- Index usage
- Join methods
- Estimated number of rows
- Filtering operations
- Query execution cost

Comparing execution plans helps in understanding query performance and identifying inefficient queries. An execution plan with fewer scanned rows or an appropriate index access path may indicate a more efficient execution strategy, although the exact interpretation depends on the database system and data distribution.

SOURCE CODE

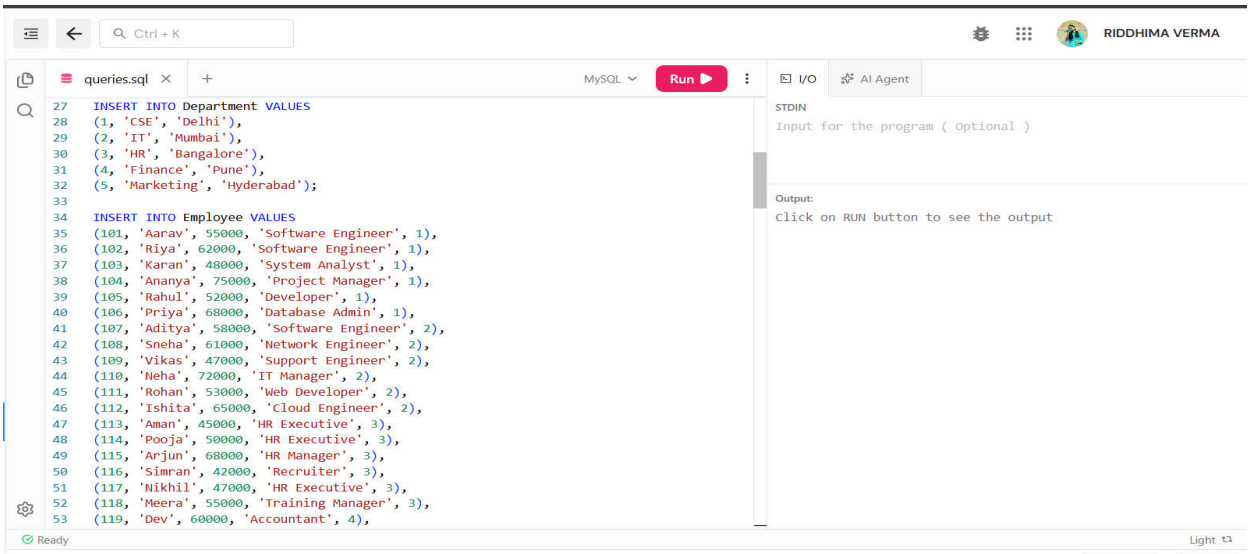



```sql
CREATE DATABASE CompanyDB;
USE CompanyDB;

CREATE TABLE Department (
    DeptID INT PRIMARY KEY,
    DeptName VARCHAR(50),
    Location VARCHAR(50)
);

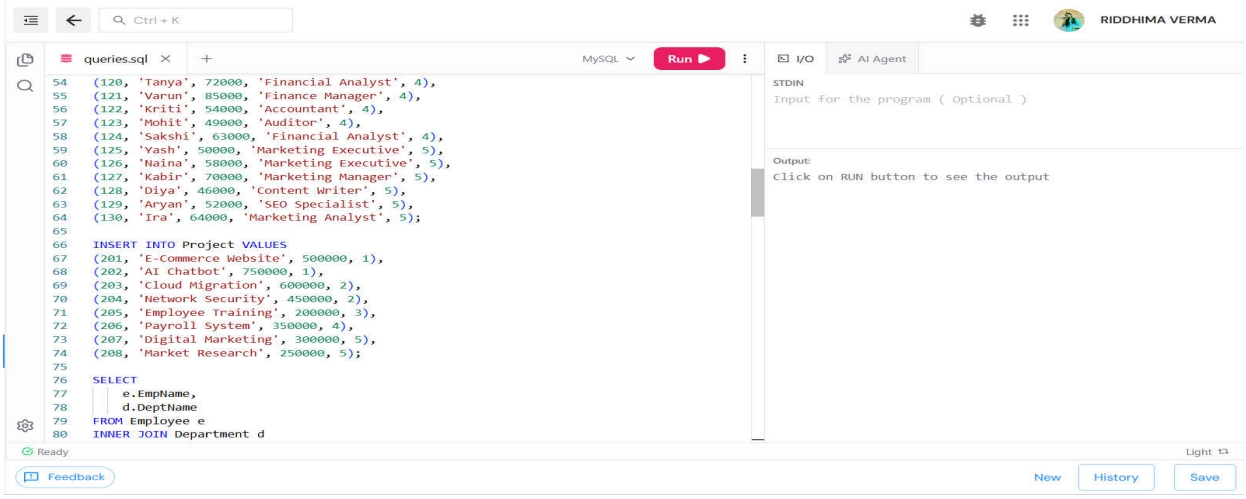
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    Salary DECIMAL(10,2),
    JobTitle VARCHAR(50),
    DeptID INT,
    FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
);

CREATE TABLE Project (
    ProjectID INT PRIMARY KEY,
    ProjectName VARCHAR(100),
    Budget DECIMAL(12,2),
    DeptID INT,
    FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
);
```




```sql
INSERT INTO Department VALUES
(1, 'CSE', 'Delhi'),
(2, 'IT', 'Mumbai'),
(3, 'HR', 'Bangalore'),
(4, 'Finance', 'Pune'),
(5, 'Marketing', 'Hyderabad');

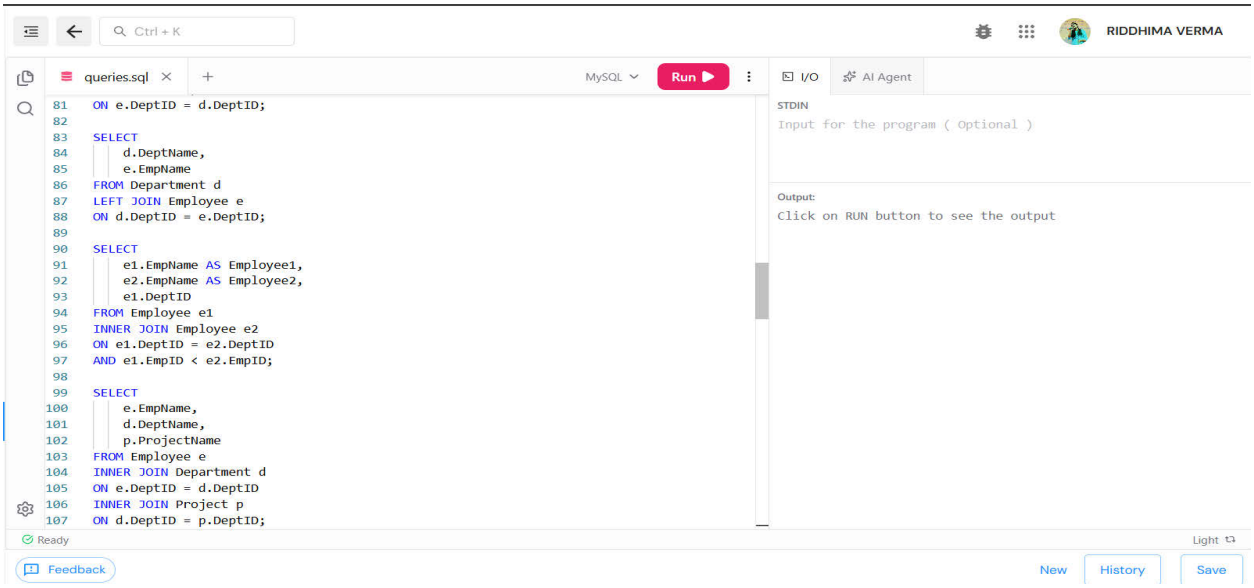
INSERT INTO Employee VALUES
(101, 'Aarav', 55000, 'Software Engineer', 1),
(102, 'Riya', 62000, 'Software Engineer', 1),
(103, 'Karan', 48000, 'System Analyst', 1),
(104, 'Ananya', 75000, 'Project Manager', 1),
(105, 'Rahul', 52000, 'Developer', 1),
(106, 'Priya', 68000, 'Database Admin', 1),
(107, 'Aditya', 58000, 'Software Engineer', 2),
(108, 'Sneha', 61000, 'Network Engineer', 2),
(109, 'Vikas', 47000, 'Support Engineer', 2),
(110, 'Neha', 72000, 'IT Manager', 2),
(111, 'Rohan', 53000, 'Web Developer', 2),
(112, 'Ishita', 65000, 'Cloud Engineer', 2),
(113, 'Aman', 45000, 'HR Executive', 3),
(114, 'Pooja', 50000, 'HR Executive', 3),
(115, 'Arjun', 68000, 'HR Manager', 3),
(116, 'Simran', 42000, 'Recruiter', 3),
(117, 'Nikhil', 47000, 'HR Executive', 3),
(118, 'Meera', 55000, 'Training Manager', 3),
(119, 'Dev', 60000, 'Accountant', 4),
```

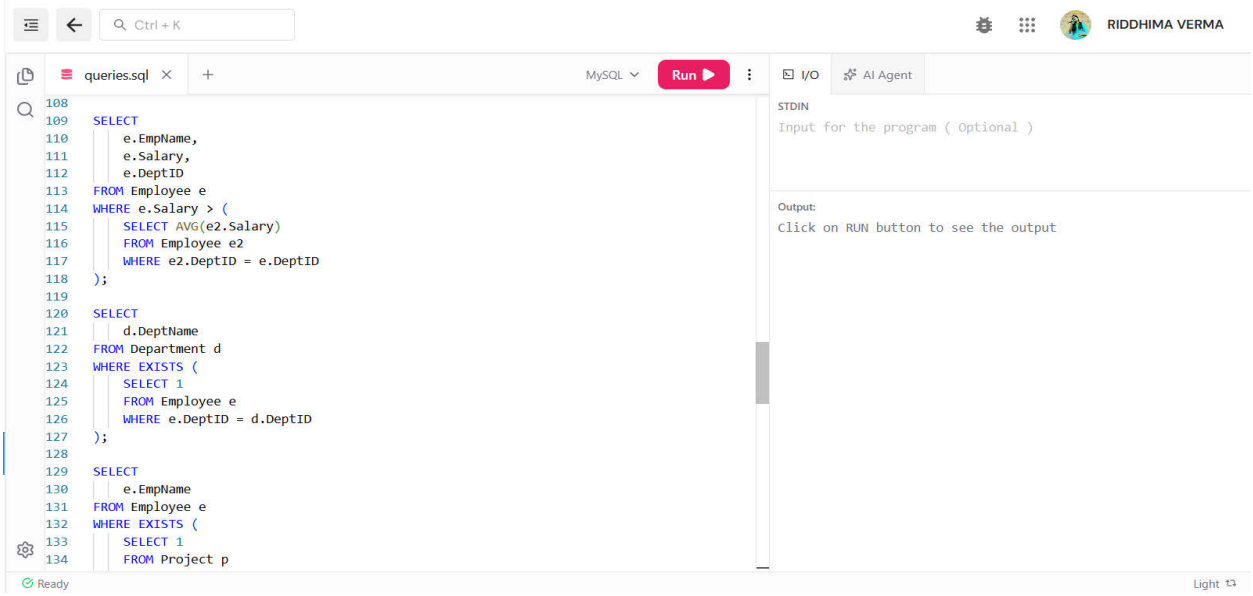


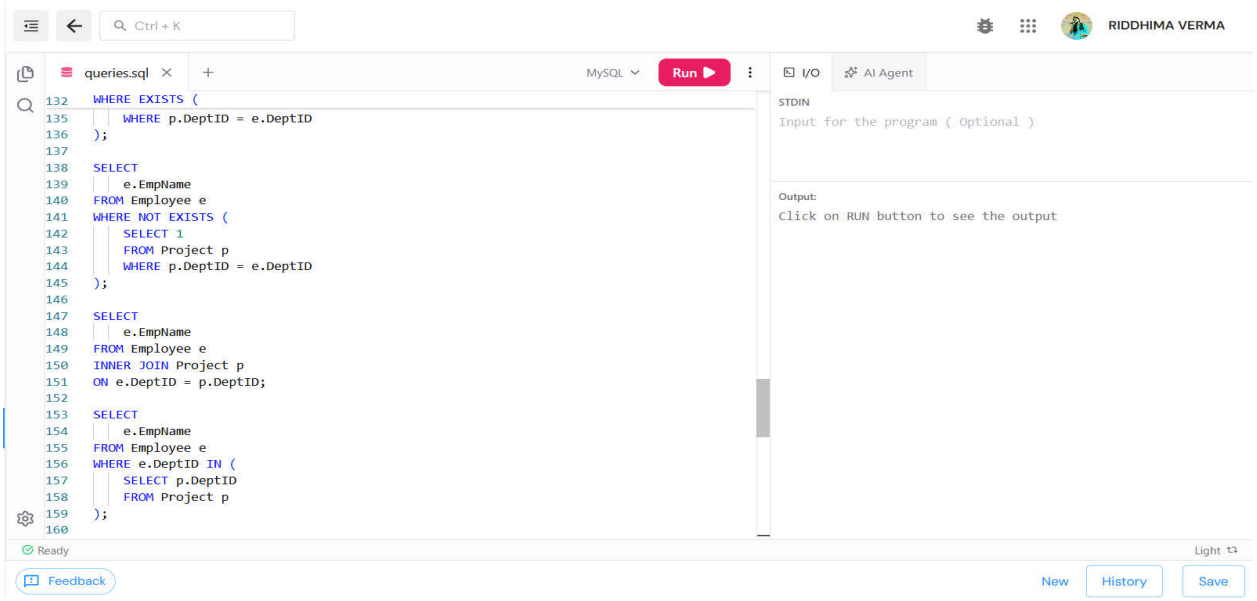

```sql
(120, 'Tanya', 72000, 'Financial Analyst', 4),
(121, 'Varun', 85000, 'Finance Manager', 4),
(122, 'Kriti', 54000, 'Accountant', 4),
(123, 'Mohit', 49000, 'Auditor', 4),
(124, 'Sakshi', 63000, 'Financial Analyst', 4),
(125, 'Yash', 50000, 'Marketing Executive', 5),
(126, 'Naina', 58000, 'Marketing Executive', 5),
(127, 'Kabir', 70000, 'Marketing Manager', 5),
(128, 'Diya', 46000, 'Content Writer', 5),
(129, 'Aryan', 52000, 'SEO Specialist', 5),
(130, 'Ira', 64000, 'Marketing Analyst', 5);

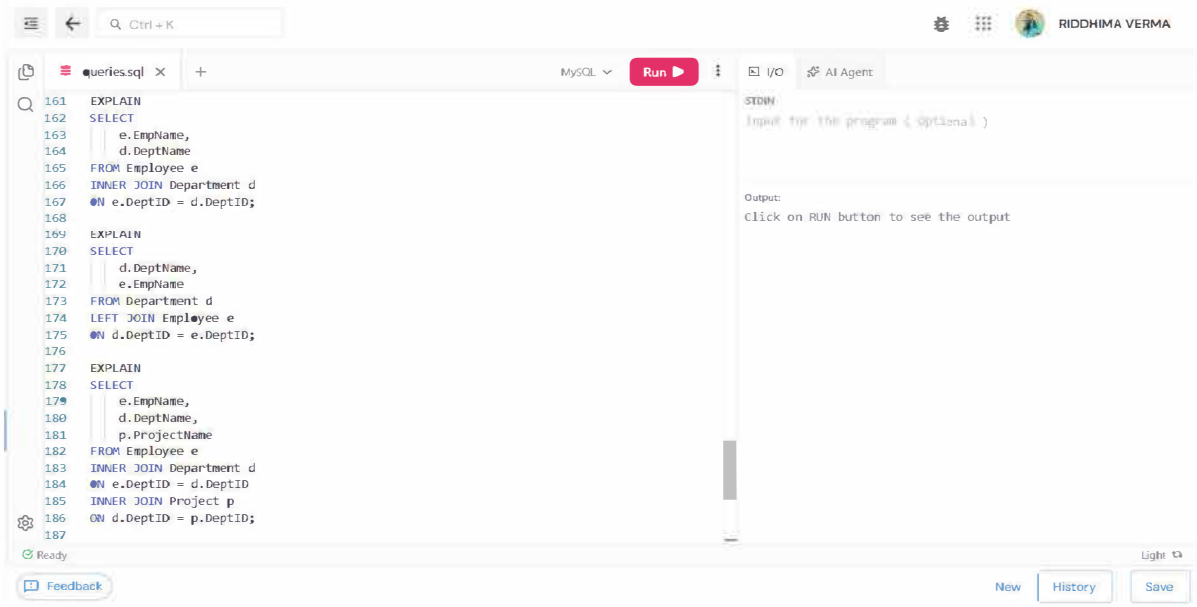
INSERT INTO Project VALUES
(201, 'E-Commerce Website', 500000, 1),
(202, 'AI Chatbot', 750000, 1),
(203, 'Cloud Migration', 600000, 2),
(204, 'Network Security', 450000, 2),
(205, 'Employee Training', 200000, 3),
(206, 'Payroll System', 350000, 4),
(207, 'Digital Marketing', 300000, 5),
(208, 'Market Research', 250000, 5);

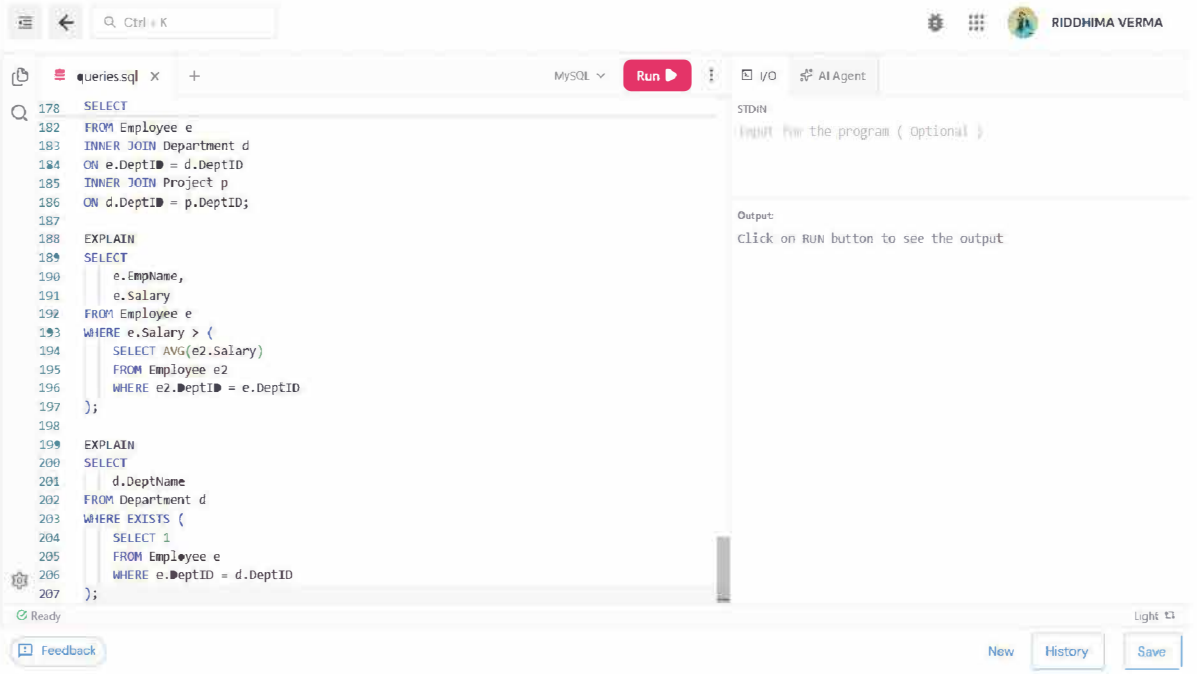
SELECT
    e.EmpName,
    d.DeptName
FROM Employee e
INNER JOIN Department d
```


Ctrl + K
RIDDHIMA VERMA
queries.sql
MySQL
Run
I/O
AI Agent
STDIN
Input for the program (Optional)
Output:
Click on RUN button to see the output
ON e.DeptID = d.DeptID;
SELECT
d.DeptName,
e.EmpName
FROM Department d
LEFT JOIN Employee e
ON d.DeptID = e.DeptID;
SELECT
e1.EmpName AS Employee1,
e2.EmpName AS Employee2,
e1.DeptID
FROM Employee e1
INNER JOIN Employee e2
ON e1.DeptID = e2.DeptID
AND e1.EmpID < e2.EmpID;
SELECT
e.EmpName,
d.DeptName,
p.ProjectName
FROM Employee e
INNER JOIN Department d
ON e.DeptID = d.DeptID
INNER JOIN Project p
ON d.DeptID = p.DeptID;
Ready
Light
Feedback
New
History
Save

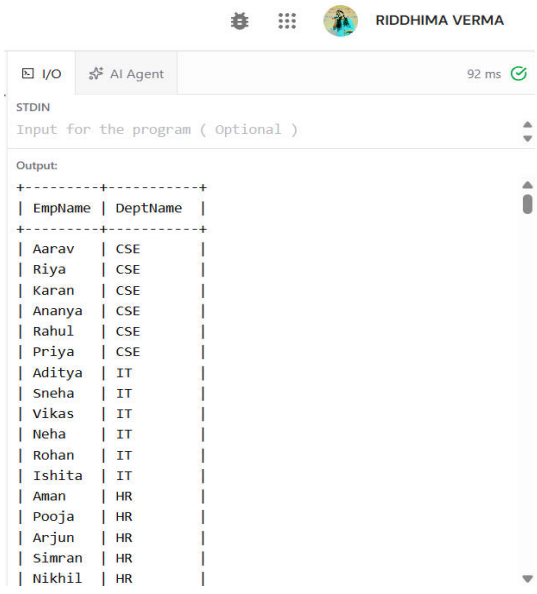

Ctrl + K
RIDDHIMA VERMA
queries.sql
MySQL
Run
I/O
AI Agent
STDIN
Input for the program (Optional)
Output:
Click on RUN button to see the output
SELECT
e.EmpName,
e.Salary,
e.DeptID
FROM Employee e
WHERE e.Salary > (
SELECT AVG(e2.Salary)
FROM Employee e2
WHERE e2.DeptID = e.DeptID
);
SELECT
d.DeptName
FROM Department d
WHERE EXISTS (
SELECT 1
FROM Employee e
WHERE e.DeptID = d.DeptID
);
SELECT
e.EmpName
FROM Employee e
WHERE EXISTS (
SELECT 1
FROM Project p
Ready
Light

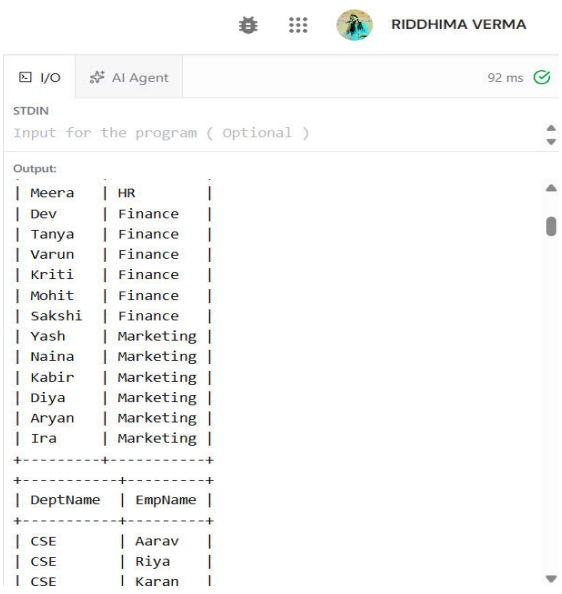

Ctrl + K
RIDDHIMA VERMA
queries.sql
MySQL
Run
I/O
AI Agent
STDIN
Input for the program (Optional)
Output:
Click on RUN button to see the output
WHERE EXISTS (
WHERE p.DeptID = e.DeptID
);
SELECT
e.EmpName
FROM Employee e
WHERE NOT EXISTS (
SELECT 1
FROM Project p
WHERE p.DeptID = e.DeptID
);
SELECT
e.EmpName
FROM Employee e
INNER JOIN Project p
ON e.DeptID = p.DeptID;
SELECT
e.EmpName
FROM Employee e
WHERE e.DeptID IN (
SELECT p.DeptID
FROM Project p
);
Ready
Light
Feedback
New
History
Save

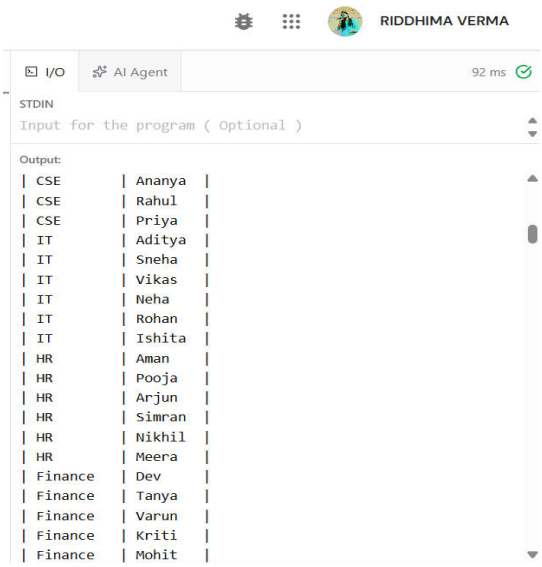

RIDDHIMA VERMA
queries.sql
MySQL
Run
I/O
AI Agent
STDIN
Input for the program (Optional)
Output:
Click on RUN button to see the output
161 EXPLAIN
162 SELECT
163 e.EmpName,
164 d.DeptName
165 FROM Employee e
166 INNER JOIN Department d
167 ON e.DeptID = d.DeptID;
168
169 EXPLAIN
170 SELECT
171 d.DeptName,
172 e.EmpName
173 FROM Department d
174 LEFT JOIN Employee e
175 ON d.DeptID = e.DeptID;
176
177 EXPLAIN
178 SELECT
179 e.EmpName,
180 d.DeptName,
181 p.ProjectName
182 FROM Employee e
183 INNER JOIN Department d
184 ON e.DeptID = d.DeptID
185 INNER JOIN Project p
186 ON d.DeptID = p.DeptID;
187
Ready
Light
Feedback
New
History
Save

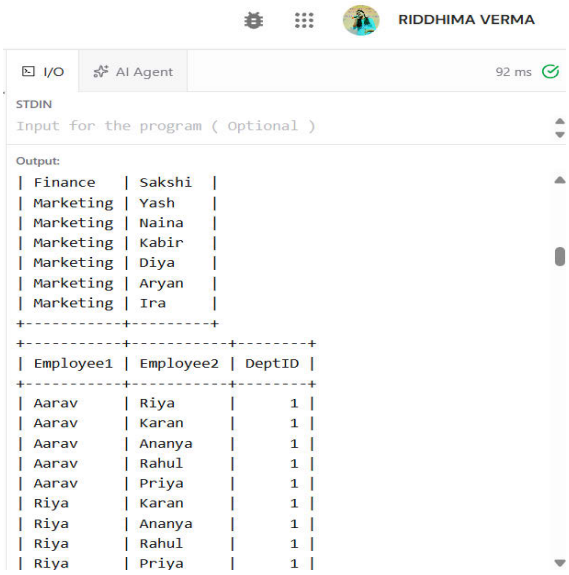

RIDDHIMA VERMA
queries.sql
MySQL
Run
I/O
AI Agent
STDIN
Input for the program (Optional)
Output:
Click on RUN button to see the output
178 SELECT
182 FROM Employee e
183 INNER JOIN Department d
184 ON e.DeptID = d.DeptID
185 INNER JOIN Project p
186 ON d.DeptID = p.DeptID;
187
188 EXPLAIN
189 SELECT
190 e.EmpName,
191 e.Salary
192 FROM Employee e
193 WHERE e.Salary > (
194 SELECT AVG(e2.Salary)
195 FROM Employee e2
196 WHERE e2.DeptID = e.DeptID
197);
198
199 EXPLAIN
200 SELECT
201 d.DeptName
202 FROM Department d
203 WHERE EXISTS (
204 SELECT 1
205 FROM Employee e
206 WHERE e.DeptID = d.DeptID
207);
Ready
Light
Feedback
New
History
Save

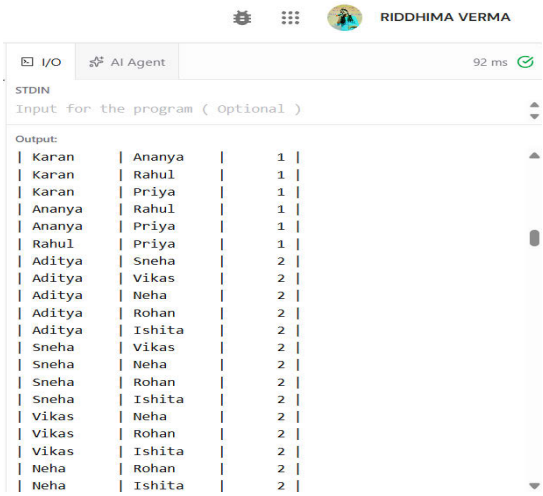
OUTPUT

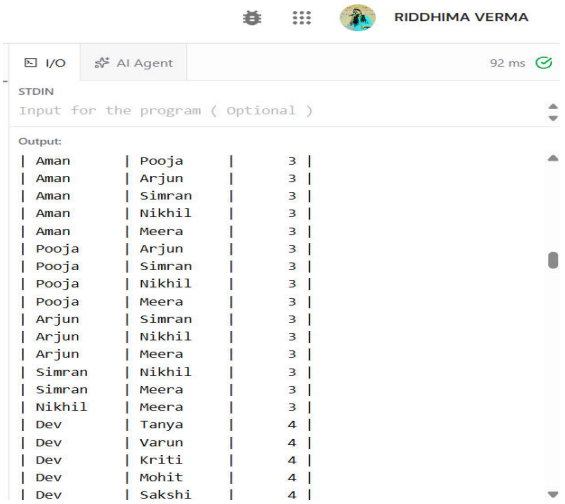

RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
+---------+-----------+
| EmpName | DeptName |
+---------+-----------+
Aarav	CSE
Riya	CSE
Karan	CSE
Ananya	CSE
Rahul	CSE
Priya	CSE
Aditya	IT
Sneha	IT
Vikas	IT
Neha	IT
Rohan	IT
Ishita	IT
Aman	HR
Pooja	HR
Arjun	HR
Simran	HR
Nikhil	HR


RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
Meera	HR
Dev	Finance
Tanya	Finance
Varun	Finance
Kriti	Finance
Mohit	Finance
Sakshi	Finance
Yash	Marketing
Naina	Marketing
Kabir	Marketing
Diya	Marketing
Aryan	Marketing
Ira	Marketing
+---------+-----------+
+-----------+---------+
| DeptName | EmpName |
+-----------+---------+
CSE	Aarav
CSE	Riya
CSE	Karan


RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
CSE	Ananya
CSE	Rahul
CSE	Priya
IT	Aditya
IT	Sneha
IT	Vikas
IT	Neha
IT	Rohan
IT	Ishita
HR	Aman
HR	Pooja
HR	Arjun
HR	Simran
HR	Nikhil
HR	Meera
Finance	Dev
Finance	Tanya
Finance	Varun
Finance	Kriti
Finance	Mohit


RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
Finance	Sakshi
Marketing	Yash
Marketing	Naina
Marketing	Kabir
Marketing	Diya
Marketing	Aryan
Marketing	Ira
+-----------+---------+
+-----------+-----------+--------+
| Employee1 | Employee2 | DeptID |
+-----------+-----------+--------+
Aarav	Riya	1
Aarav	Karan	1
Aarav	Ananya	1
Aarav	Rahul	1
Aarav	Priya	1
Riya	Karan	1
Riya	Ananya	1
Riya	Rahul	1
Riya	Priya	1


RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
Karan	Ananya	1
Karan	Rahul	1
Karan	Priya	1
Ananya	Rahul	1
Ananya	Priya	1
Rahul	Priya	1
Aditya	Sneha	2
Aditya	Vikas	2
Aditya	Neha	2
Aditya	Rohan	2
Aditya	Ishita	2
Sneha	Vikas	2
Sneha	Neha	2
Sneha	Rohan	2
Sneha	Ishita	2
Vikas	Neha	2
Vikas	Rohan	2
Vikas	Ishita	2
Neha	Rohan	2
Neha	Ishita	2


RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
Aman	Pooja	3
Aman	Arjun	3
Aman	Simran	3
Aman	Nikhil	3
Aman	Meera	3
Pooja	Arjun	3
Pooja	Simran	3
Pooja	Nikhil	3
Pooja	Meera	3
Arjun	Simran	3
Arjun	Nikhil	3
Arjun	Meera	3
Simran	Nikhil	3
Simran	Meera	3
Nikhil	Meera	3
Dev	Tanya	4
Dev	Varun	4
Dev	Kriti	4
Dev	Mohit	4
Dev	Sakshi	4

RIDDHIMA VERMA

I/O | AI Agent | 92 ms

STDIN

Input for the program (Optional)

Output:

```
| Tanya    | Varun    |      4 |
| Tanya    | Kriti    |      4 |
| Tanya    | Mohit    |      4 |
| Tanya    | Sakshi   |      4 |
| Varun    | Kriti    |      4 |
| Varun    | Mohit    |      4 |
| Varun    | Sakshi   |      4 |
| Kriti    | Mohit    |      4 |
| Kriti    | Sakshi   |      4 |
| Mohit    | Sakshi   |      4 |
| Yash     | Naina    |      5 |
| Yash     | Kabir    |      5 |
| Yash     | Diya     |      5 |
| Yash     | Aryan    |      5 |
| Yash     | Ira      |      5 |
| Naina    | Kabir    |      5 |
| Naina    | Diya     |      5 |
| Naina    | Aryan    |      5 |
| Naina    | Ira      |      5 |
| Kabir    | Diya     |      5 |
```

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I/O | AI Agent | 92 ms

STDIN

Input for the program (Optional)

Output:

```
| Kabir    | Aryan    |      5 |
| Kabir    | Ira      |      5 |
| Diya     | Aryan    |      5 |
| Diya     | Ira      |      5 |
| Aryan    | Ira      |      5 |
+----------+----------+--------+
+---------+-----------+--------------------+
| EmpName | DeptName  | ProjectName        |
+---------+-----------+--------------------+
| Aarav   | CSE       | E-Commerce Website |
| Riya    | CSE       | E-Commerce Website |
| Karan   | CSE       | E-Commerce Website |
| Ananya  | CSE       | E-Commerce Website |
| Rahul   | CSE       | E-Commerce Website |
| Priya   | CSE       | E-Commerce Website |
| Aarav   | CSE       | AI Chatbot         |
| Riya    | CSE       | AI Chatbot         |
| Karan   | CSE       | AI Chatbot         |
| Ananya  | CSE       | AI Chatbot         |
| Rahul   | CSE       | AI Chatbot         |
```

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I/O | AI Agent | 92 ms

STDIN

Input for the program (Optional)

Output:

```
| Priya    | CSE         | AI Chatbot          |
| Aditya   | IT          | Cloud Migration     |
| Sneha    | IT          | Cloud Migration     |
| Vikas    | IT          | Cloud Migration     |
| Neha     | IT          | Cloud Migration     |
| Rohan    | IT          | Cloud Migration     |
| Ishita   | IT          | Cloud Migration     |
| Aditya   | IT          | Network Security    |
| Sneha    | IT          | Network Security    |
| Vikas    | IT          | Network Security    |
| Neha     | IT          | Network Security    |
| Rohan    | IT          | Network Security    |
| Ishita   | IT          | Network Security    |
| Aman     | HR          | Employee Training   |
| Pooja    | HR          | Employee Training   |
| Arjun    | HR          | Employee Training   |
| Simran   | HR          | Employee Training   |
| Nikhil   | HR          | Employee Training   |
| Meera    | HR          | Employee Training   |
| Dev      | Finance     | Payroll System      |
```

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I/O | AI Agent | 92 ms

STDIN

Input for the program (Optional)

Output:

```
| Tanya    | Finance    | Payroll System     |
| Varun    | Finance    | Payroll System     |
| Kriti    | Finance    | Payroll System     |
| Mohit    | Finance    | Payroll System     |
| Sakshi   | Finance    | Payroll System     |
| Yash     | Marketing  | Digital Marketing  |
| Naina    | Marketing  | Digital Marketing  |
| Kabir    | Marketing  | Digital Marketing  |
| Diya     | Marketing  | Digital Marketing  |
| Aryan    | Marketing  | Digital Marketing  |
| Ira      | Marketing  | Digital Marketing  |
| Yash     | Marketing  | Market Research    |
| Naina    | Marketing  | Market Research    |
| Kabir    | Marketing  | Market Research    |
| Diya     | Marketing  | Market Research    |
| Aryan    | Marketing  | Market Research    |
| Ira      | Marketing  | Market Research    |
+----------+------------+--------------------+
+---------+----------+--------+
| EmpName | Salary   | DeptID |
```

RIDDHIMA VERMA

I/O | AI Agent | 92 ms

STDIN

Input for the program (Optional)

Output:

```
| EmpName | Salary   | DeptID |
+---------+----------+--------+
| Riya    | 62000.00 |      1 |
| Ananya  | 75000.00 |      1 |
| Priya   | 68000.00 |      1 |
| Sneha   | 61000.00 |      2 |
| Neha    | 72000.00 |      2 |
| Ishita  | 65000.00 |      2 |
| Arjun   | 68000.00 |      3 |
| Meera   | 55000.00 |      3 |
| Tanya   | 72000.00 |      4 |
| Varun   | 85000.00 |      4 |
| Naina   | 58000.00 |      5 |
| Kabir   | 70000.00 |      5 |
| Ira     | 64000.00 |      5 |
+---------+----------+--------+
+------------+
| DeptName   |
+------------+
| CSE        |
```

RIDDHIMA VERMA

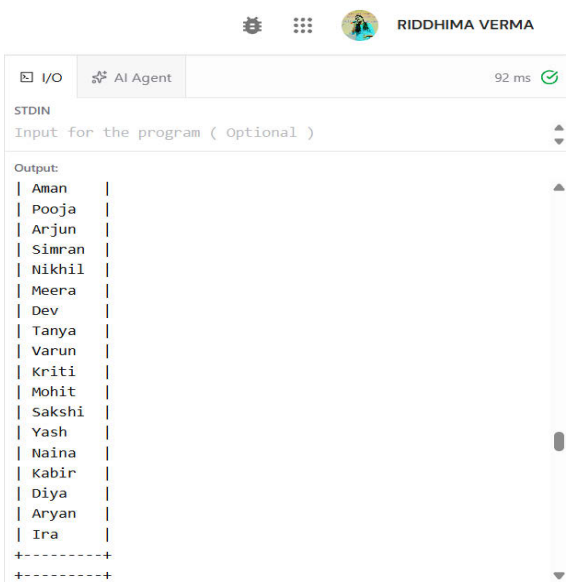
I/O | AI Agent | 92 ms

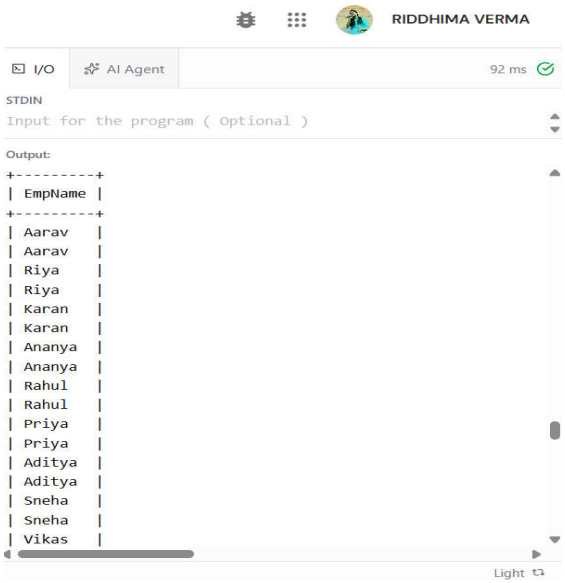
STDIN

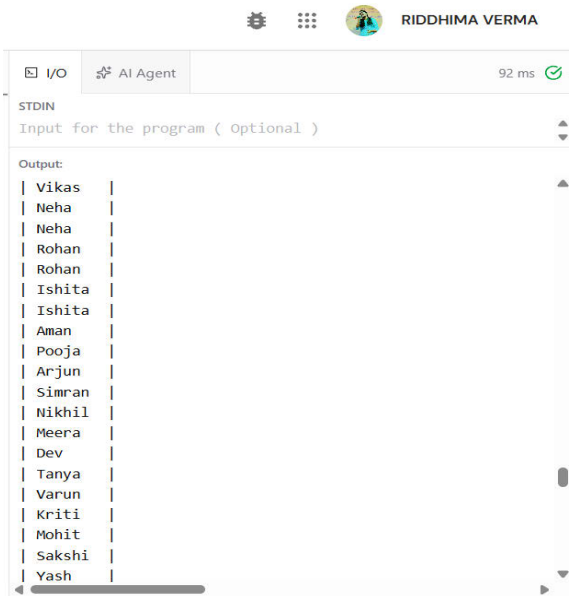
Input for the program (Optional)

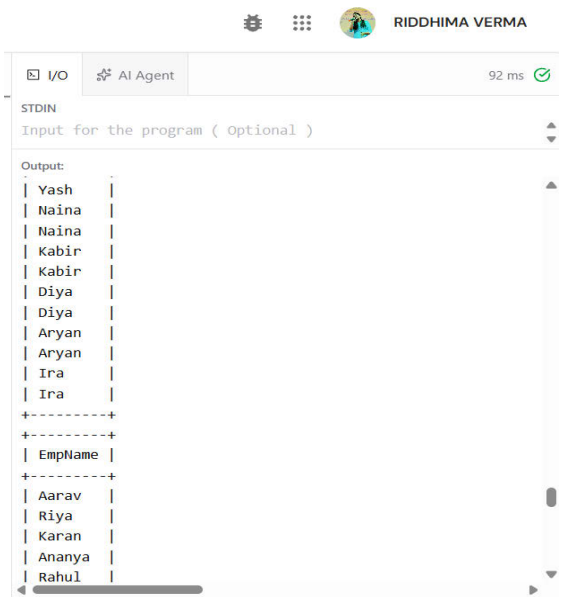
Output:

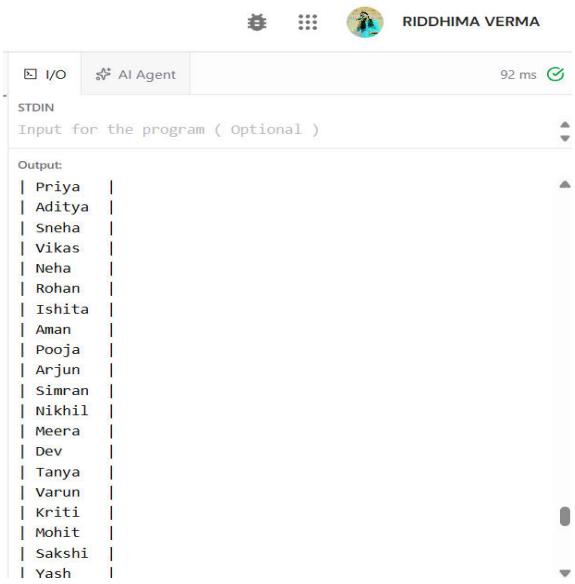
```
| IT          |
| HR          |
| Finance     |
| Marketing   |
+-------------+
+----------+
| EmpName  |
+----------+
| Aarav    |
| Riya     |
| Karan    |
| Ananya   |
| Rahul    |
| Priya    |
| Aditya   |
| Sneha    |
| Vikas    |
| Neha     |
| Rohan    |
| Ishita   |
```


RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
| Aman |
| Pooja |
| Arjun |
| Simran |
| Nikhil |
| Meera |
| Dev |
| Tanya |
| Varun |
| Kriti |
| Mohit |
| Sakshi |
| Yash |
| Naina |
| Kabir |
| Diya |
| Aryan |
| Ira |


RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
+---------+
| EmpName |
+---------+
| Aarav |
| Aarav |
| Riya |
| Riya |
| Karan |
| Karan |
| Ananya |
| Ananya |
| Rahul |
| Rahul |
| Priya |
| Priya |
| Aditya |
| Aditya |
| Sneha |
| Sneha |
| Vikas |
Light


RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
| Vikas |
| Neha |
| Neha |
| Rohan |
| Rohan |
| Ishita |
| Ishita |
| Aman |
| Pooja |
| Arjun |
| Simran |
| Nikhil |
| Meera |
| Dev |
| Tanya |
| Varun |
| Kriti |
| Mohit |
| Sakshi |
| Yash |


RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
| Yash |
| Naina |
| Naina |
| Kabir |
| Kabir |
| Diya |
| Diya |
| Aryan |
| Aryan |
| Ira |
| Ira |
+---------+
+---------+
| EmpName |
+---------+
| Aarav |
| Riya |
| Karan |
| Ananya |
| Rahul |


RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
| Priya |
| Aditya |
| Sneha |
| Vikas |
| Neha |
| Rohan |
| Ishita |
| Aman |
| Pooja |
| Arjun |
| Simran |
| Nikhil |
| Meera |
| Dev |
| Tanya |
| Varun |
| Kriti |
| Mohit |
| Sakshi |
| Yash |


Ctrl + K
RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
| Naina |
| Kabir |
| Diya |
| Aryan |
| Ira |
+---------+
id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	NULL	ALL	PRIMARY	NULL	NULL	NULL	5	100.00	NULL
1	SIMPLE	e	NULL	ref	DeptID	DeptID	5	sandbox_db.d.DeptID	15	100.00	NULL
id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	NULL	ALL	NULL	NULL	NULL	NULL	5	100.00	NULL
1	SIMPLE	e	NULL	ref	DeptID	DeptID	5	sandbox_db.d.DeptID	15	100.00	NULL
id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra


Ctrl + K
RIDDHIMA VERMA
I/O
AI Agent
92 ms
STDIN
Input for the program (Optional)
Output:
SIMPLE	d	NULL	ALL	NULL	NULL	NULL	NULL	5	100.00	NULL
SIMPLE	e	NULL	ref	DeptID	DeptID	5	sandbox_db.d.DeptID	15	100.00	NULL
select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
SIMPLE	d	NULL	ALL	PRIMARY	NULL	NULL	NULL	5	100.00	NULL
SIMPLE	p	NULL	ref	DeptID	DeptID	5	sandbox_db.d.DeptID	1	100.00	NULL
SIMPLE	e	NULL	ref	DeptID	DeptID	5	sandbox_db.d.DeptID	15	100.00	NULL
select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
PRIMARY	e	NULL	ALL	NULL	NULL	NULL	NULL	30	100.00	Using where
DEPENDENT SUBQUERY	e2	NULL	ref	DeptID	DeptID	5	sandbox_db.e.DeptID	15	100.00	NULL
select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
SIMPLE	e	NULL	index	DeptID	DeptID	5	NULL	30	6.67	Using where; Using index; LooseScan
SIMPLE	d	NULL	eq_ref	PRIMARY	PRIMARY	4	sandbox_db.e.DeptID	1	100.00	NULL

OBSERVATIONS/OUTCOME

1. Employee, Department, and Project tables were successfully created and populated with the required data.
2. INNER JOIN successfully retrieved matching employee and department records.
3. LEFT JOIN successfully displayed all departments along with their corresponding employees.
4. A self-join successfully identified employees belonging to the same department.
5. A 3-way join successfully retrieved employee, department, and project details together.
6. The correlated subquery successfully identified employees whose salary was greater than the average salary of their respective department.
7. The EXISTS operator successfully checked for the existence of projects associated with employees' departments.
8. INTERSECT and EXCEPT operations were successfully simulated using EXISTS and NOT EXISTS.
9. EXPLAIN successfully displayed the execution plans for different SQL queries, allowing their execution strategies to be examined and compared.